

MAYANK KUMAR

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Email

LinkedIn

GitHub

Portfolio

LeetCode

Profile Summary

- Full Stack Developer with 3.3+ years building scalable web applications using React, Node.js, Next.js, and AWS. Expert in AI Agent development with RAG architectures and LLM integration. Proven fintech experience with microservices and serverless architectures.

Technical Skills

Full Stack Development: JavaScript, TypeScript, Python, Node.js, React, Next.js, Express.js, Redux, Recoil, MERN Stack

AWS Cloud Services: Lambda, API Gateway, EC2, S3, SQS, RDS, ECS, CDK, Cognito, CloudWatch, Textract, Bedrock

AI Agents & Machine Learning: RAG Architectures, Vector Databases, LLMs, AI Workflows, Intelligent Automation

Databases & Storage: MongoDB, MySQL, PostgreSQL, DynamoDB, Vector Databases, SQL

DevOps & Tools: Docker, Microservices Architecture, RESTful APIs, OAuth 2.0, Material UI, Tailwind CSS, ShadCN

Core Competencies: System Design, Data Structures & Algorithms, Agile Methodologies, Problem Solving

Experience

SMC Global Securities — Fintech

July 2024 – Present

Member of Technical Staff | Full Stack Developer

Delhi, India

- Built **AI Agent sentiment analysis system** with **RAG pipeline** using **AWS Bedrock Claude 3.7 Sonnet** and vector embeddings, improving customer insights by 40%.
- Migrated legacy Windows Forms **RegTech App** to **Next.js** with serverless AWS architecture (**S3, SQS, RDS, Lambda**), Oracle to SQL Server migration.
- Developed **URL Shortener service** with React dashboard using **AWS CDK, Lambda, API Gateway, Cognito** with OAuth 2.0 authentication.
- Built **AI signature verification system** with **RAG architecture** using **Next.js, AWS Textract, DynamoDB, Llama 3.2 Vision**, achieving 86% accuracy.

Infosys Limited

June 2022 – July 2024

Digital Specialist Engineer | Full Stack Developer

Chandigarh, Haryana

- Built full stack railcar tracking application for **Wabtec Corporation** using **Node.js, React.js**, enhancing real-time shipment visibility and inventory control.
- Designed **RESTful APIs** using **Node.js, Express.js** for real-time container tracking, processing 10,000+ daily requests with 99.9% uptime.
- Implemented **Cerbos RBAC/ABAC** authorization in **microservices architecture**, reducing security incidents by 75%.
- Optimized frontend performance by 40% using **React.js, Material UI, Recoil**, achieving 30% faster page loads.

Projects

Pre-Order Service Project Revamp | Next.js, TypeScript, AWS CDK, Node.js, Microservices

- Migrated legacy Windows Forms to **Next.js** trade order system with TypeScript, Oracle to SQL Server migration for 60% performance improvement.
- Built serverless voice processing pipeline using **AWS CDK, S3, SQS, ECS, RDS PostgreSQL** with event-driven architecture.
- Developed containerized Node.js microservice for voice metadata extraction with resilient architecture and auto-scaling.

AI Agent Chat Sentiment Analysis Platform | AWS Bedrock, Claude 3.7 Sonnet, Lambda, Next.js

- Built **AI Agent chat analysis platform** with **RAG-powered** context retrieval using AWS Bedrock Claude 3.7 Sonnet and serverless Lambda functions.
- Developed Next.js dashboard for real-time sentiment, entity extraction and analytics with DynamoDB integration.

AI Agent Signature Verification System | Next.js, AWS Textract, Bedrock, Llama 3.2 Vision

- Built production **AI Agent signature verification system** with **RAG architecture** using AWS Textract, Next.js, DynamoDB, and Llama 3.2 Vision.
- Implemented **RAG-enhanced workflows** with Llama 3.2 Vision on AWS Bedrock for signature validation and compliance verification with 86% accuracy.

Education Achievements

Rajkiya Engineering College, Azamgarh

Aug. 2018 – May 2022

Bachelor of Technology in Information Technology

Uttar Pradesh

* Solved 350+ DSA problems on **LeetCode**, **CodeChef**, **GFG**, **HackerRank**. 3 Star CodeChef coder (1551 rating).